

WEIGHTED CERTIFICATES OVER REAL AND FINITE FIELDS FOR GENERALIZED PETERSEN GRAPHS

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ABSTRACT. We study the corrected conjecture $Z(P(n, 3)) = 8$ for $n \geq 13$ using weighted graph-pattern matrices. Over the reals, Fourier certificates prove $M(P(n, 3)) = Z(P(n, 3)) = 8$ whenever 10, 24, 28, or 42 divides n . We retain the complete six-triple rational classification and strengthen the arithmetic obstruction: no prime-power level supports nullity eight in the real constant-weight symmetric class. An exact modular sieve followed by cyclotomic verification gives a complete classification through level 420.

Changing the coefficient field yields additional zero-forcing families. A polynomial greatest-common-divisor formula computes the nullity of a five-parameter cyclic matrix over any field, including when the characteristic divides the graph order. Explicit finite-field certificates prove $Z(P(Lr, 3)) = 8$ for all $r \geq 1$ and $L \in \{16, 17, 18, 19, 21, 22, 23, 25, 26, 27, 29, 31\}$. In particular, a binary degree-eight divisor of $z^{17} - 1$ proves the entire family $Z(P(17r, 3)) = 8$. These certificates do not assert real maximum nullity eight.

For the signed real specialization, we determine every root-of-unity zero and prove that $Q_k(z) = z^{2k+2} + z^{2k} + z^{k+1} + z^2 + 1$ is squarefree cyclotomic exactly when $k = 2, 3, 7$. Finally, a first-force anchoring lemma reduces exhaustive seven-vertex zero-forcing search to $O(n^4)$ candidate sets and verifies the conjecture for every $13 \leq n \leq 64$. The uniform conjecture remains open.

1. INTRODUCTION

Let G be a finite simple graph. A set of initially blue vertices is a *zero forcing set* if repeated application of the colour-change rule—a blue vertex with exactly one white neighbour forces that neighbour blue—eventually colours every vertex blue. The minimum cardinality of such a set is the *zero forcing number* $Z(G)$. Zero forcing was introduced in connection with the minimum-rank problem for graph-described symmetric matrices, and the fundamental inequality

$$M(G) \leq Z(G) \tag{1}$$

relates it to the maximum nullity $M(G)$; see [1] and the surveys [6, 3, 14].

For integers $n \geq 3$ and $1 \leq k < n/2$, the *generalized Petersen graph* $P(n, k)$ has outer vertices $u_0, \dots, u_{n-1}$, inner vertices $v_0, \dots, v_{n-1}$, and edges

$$u_i u_{i+1}, \quad u_i v_i, \quad v_i v_{i+k},$$

subscripts modulo n . These graphs originate with Watkins [13]; their ordinary adjacency spectrum was described by Gera and Stănică [7], who already used the 2×2 Fourier block decomposition that we generalize below.

Alameda et al. [2] proved the general upper bound

$$Z(P(n, k)) \leq 2k + 2 \tag{2}$$

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for every admissible n, k , and used high-multiplicity eigenvalues of *ordinary* adjacency matrices to establish

$$M(P(15r, 2)) = Z(P(15r, 2)) = 6, \quad M(P(24r, 5)) = Z(P(24r, 5)) = 12$$

for every $r \geq 1$. Rashidi, Shajareh Poursalavati and Tavakkoli [11] subsequently claimed the value 8 for $Z(P(n, 3))$ whenever $n \geq 12$. Krishnan [8] identified the boundary counterexample $Z(P(12, 3)) = 7$, established the upper bound $Z(P(n, 3)) \leq 8$ for $n \geq 9$, computed the zero forcing numbers for $7 \leq n \leq 20$, and formulated the corrected conjecture

$$Z(P(n, 3)) = 8 \quad (n \geq 13). \quad (3)$$

The unresolved content of (3) is therefore the uniform lower bound $Z(P(n, 3)) \geq 8$. Our independent computation in Section 9 reproduces Krishnan's values through $n = 20$ and extends the exhaustive verification of (3) through $n = 64$, using the anchoring reduction proved there.

The present paper attacks that lower bound through real maximum nullity and through kernel dimensions over finite fields. It also studies rigorous limitations of the real certificate class. The basic idea enlarges the adjacency-matrix certificate of [2]: instead of searching only for a high-multiplicity eigenvalue of the ordinary adjacency matrix, we choose nonzero edge weights, diagonal shifts, and Fourier modes simultaneously. This produces matrices in $\mathcal{S}(P(n, k))$ whose kernels can be certified mode by mode. Related maximum-nullity and zero-forcing questions for circulant graph families are studied by Duong et al. [5]. Here, instead, a 2×2 block-circulant matrix for generalized Petersen graphs is designed so that prescribed Fourier modes are singular through a reciprocal polynomial.

Results. Our first result is the positive one.

Theorem 1.1. *If n is divisible by at least one of 10, 24, 28, 42, then*

$$M(P(n, 3)) = Z(P(n, 3)) = 8.$$

Consequently the value in (3) holds on a set of indices of natural density $1/6$.

The proof is explicit: for each modulus we construct a fixed weighting whose Fourier determinant vanishes at eight prescribed roots of unity. The remaining results delimit the method.

Theorem 1.2 (Completeness over $\mathbb{Q}$; Theorem 5.1). *Let $p, q, \tau \in \mathbb{Q}$ with $\tau \neq 0$, and suppose $R_{p,q,\tau}$ of (7) has eight distinct roots, all roots of unity. Then (p, q, τ) is one of the six triples of Table 2, and the modulus L is 10, 24, 40 or 42.*

Theorem 1.3 (Arithmetic obstructions; Theorems 6.2, 6.3, 6.4). *Suppose $A_n(p, q, \tau)$ has nullity 8 for some real p, q and $\tau \neq 0$. Then 27 lies in the numerical semigroup generated by the primes dividing n ; in particular n is not a power of 2, and n has an odd prime divisor at most 23. Moreover n is not prime, and if $n = 2p$ with p an odd prime then $p = 5$.*

The prime obstruction already shows that no collection of real symmetric constant-weight block-circulant certificates of this form—finite or infinite—can prove (3) for all sufficiently large n . A computer-assisted search reported in Section 6.5 finds no additional minimal modulus through $n = 420$ beyond 10, 24, 28, 42. The exact sieve now proves completeness within this finite range, but does not give a global classification.

Our second positive result gives a reusable signed construction for arbitrary k .

Theorem 1.4. *Let $Q_k(z) = z^{2k+2} + z^{2k} + z^{k+1} + z^2 + 1$. If $Q_k(z)$ divides $z^L - 1$ in $\mathbb{Z}[z]$, then for every integer $r \geq 1$,*

$$M(P(Lr, k)) = Z(P(Lr, k)) = 2k + 2.$$

Since $Q_7(z) = \Phi_5(z)\Phi_{10}(z)\Phi_{24}(z)$, Theorem 1.4 yields the exact family $M(P(120r, 7)) = Z(P(120r, 7)) = 16$. Here and below Φ_m denotes the m th cyclotomic polynomial.

The distinction between the coefficient fields is consequential. The prime-power obstruction concerns real constant-weight symmetric matrices; it does not obstruct finite-field kernels. Section 8 proves the following additional result.

Theorem 1.5. *For every positive integer r and every base level*

$$L \in \{16, 17, 18, 19, 21, 22, 23, 25, 26, 27, 29, 31\},$$

we have

$$Z(P(Lr, 3)) = 8.$$

The proof supplies fixed finite-field matrices whose nullity is eight at all multiples of the listed base levels. No conclusion about $M(P(Lr, 3))$ over $\mathbb{R}$ follows from this argument alone. The especially small certificate at $L = 17$ uses only coefficients in $\mathbb{F}_2$. We also resolve the signed classification question exactly, excluding every k other than 2, 3, 7, and exclude all prime powers from the real class.

Organization. Section 2 develops the weighted Fourier certificate and the reciprocal polynomial for $P(n, 3)$. Section 3 gives the mode-selection criterion and a symmetry. Section 4 proves the explicit families and the density statement. Section 5 classifies all rational certificates. Section 6 proves the arithmetic obstructions and reports a computer-assisted finite search. Section 7 proves the signed criterion and classifies all its cyclotomic cases. Section 8 develops certificates over finite fields. Section 9 records the independent computation of $Z(P(n, 3))$. Section 10 discusses limitations and open problems.

2. WEIGHTED FOURIER CERTIFICATES

For a graph G on N vertices let $\mathcal{S}(G)$ be the set of real symmetric $N \times N$ matrices $A = [a_{ij}]$ such that, for $i \neq j$,

$$a_{ij} \neq 0 \iff ij \in E(G),$$

with arbitrary diagonal entries. Thus $M(G) = \max\{\text{null } A : A \in \mathcal{S}(G)\}$.

Let C_n be the cyclic shift defined by $C_n e_i = e_{i+1}$, indices modulo n . Put $\omega = e^{2\pi i/n}$ and

$$f_j = (1, \omega^{-j}, \omega^{-2j}, \dots, \omega^{-(n-1)j})^T, \quad 0 \leq j < n.$$

Then $C_n f_j = \omega^j f_j$. After normalization the f_j form an orthonormal Fourier basis, so every polynomial in C_n is unitarily diagonalized by that basis [4].

Lemma 2.1 (Fourier block decomposition). *Let $a, b, c \in \mathbb{R} \setminus \{0\}$ and $d, e \in \mathbb{R}$. Define*

$$H_{n,k}(a, b, c, d, e) = \begin{pmatrix} dI_n + a(C_n + C_n^{-1}) & bI_n \\ bI_n & eI_n + c(C_n^k + C_n^{-k}) \end{pmatrix}. \quad (4)$$

Then $H_{n,k}(a, b, c, d, e) \in \mathcal{S}(P(n, k))$. Over $\mathbb{C}$ it is unitarily similar to the direct sum of the 2×2 matrices

$$B_j = \begin{pmatrix} d + a(\omega^j + \omega^{-j}) & b \\ b & e + c(\omega^{kj} + \omega^{-kj}) \end{pmatrix}, \quad 0 \leq j < n, \quad (5)$$

and consequently $\text{null } H_{n,k}(a, b, c, d, e) = \sum_{j=0}^{n-1} \text{null } B_j$.

Proof. Since $1 \leq k < n/2$ we have $C_n^k \neq C_n^{-k}$, so the nonzero weights a, b, c in (4) occur precisely on the outer, spoke and inner edges of $P(n, k)$ respectively, and the off-diagonal zero pattern is correct. The vectors f_j simultaneously diagonalize C_n and all its powers; restricting (4) to the span of $(f_j, 0)^T$ and $(0, f_j)^T$ gives (5). A real matrix has the same rank over $\mathbb{R}$ and over $\mathbb{C}$, so the direct-sum decomposition computes its real nullity. $\square$

2.1. A normalized three-parameter family. For $p, q \in \mathbb{R}$ and $\tau \in \mathbb{R} \setminus \{0\}$ set $\varepsilon = \text{sgn}(\tau)$, $\beta = \sqrt{|\tau|}$, and define

$$A_n(p, q, \tau) = \begin{pmatrix} pI_n + C_n + C_n^{-1} & \beta I_n \\ \beta I_n & \varepsilon(qI_n + C_n^3 + C_n^{-3}) \end{pmatrix}. \quad (6)$$

Since the three edge weights $1, \beta, \varepsilon$ are nonzero, $A_n(p, q, \tau) \in \mathcal{S}(P(n, 3))$. Define the reciprocal polynomial

$$R_{p,q,\tau}(z) = z^8 + pz^7 + z^6 + qz^5 + (pq - \tau)z^4 + qz^3 + z^2 + pz + 1. \quad (7)$$

Remark 2.2. The family (6) captures the singular-mode behavior of every constant-weight block-circulant matrix (4) for $k = 3$. Indeed, the Fourier determinant of (4) is

$$(d + ax)(e + cy) - b^2 = ac \left[\left(\frac{d}{a} + x \right) \left(\frac{e}{c} + y \right) - \frac{b^2}{ac} \right],$$

where $x = z + z^{-1}$ and $y = z^3 + z^{-3}$. Thus, with

$$p = d/a, \quad q = e/c, \quad \tau = b^2/(ac),$$

the general block is singular for exactly the same Fourier modes as the block of $A_n(p, q, \tau)$. Since both off-diagonal entries are nonzero, every singular 2×2 block has nullity one. Hence the two matrices have the same nullity. Consequently the obstructions in Section 6 apply to the entire constant-weight block-circulant class (4), not merely to a special normalization.

Lemma 2.3. *Let $z = \omega^j$ be an n th root of unity and let B_j be the Fourier block of $A_n(p, q, \tau)$. Then*

$$\det B_j = \varepsilon z^{-4} R_{p,q,\tau}(z). \quad (8)$$

Hence B_j is singular if and only if $R_{p,q,\tau}(z) = 0$, and every singular block has nullity exactly one.

Proof. By Lemma 2.1, $B_j = \begin{pmatrix} p + z + z^{-1} & \beta \\ \beta & \varepsilon(q + z^3 + z^{-3}) \end{pmatrix}$. Because $\beta^2 = |\tau| = \varepsilon\tau$, $\det B_j = \varepsilon[(p + z + z^{-1})(q + z^3 + z^{-3}) - \tau]$. Writing $p + z + z^{-1} = z^{-1}(z^2 + pz + 1)$ and $q + z^3 + z^{-3} = z^{-3}(z^6 + qz^3 + 1)$ and multiplying by z^4 gives

$$z^4 \varepsilon^{-1} \det B_j = (z^2 + pz + 1)(z^6 + qz^3 + 1) - \tau z^4 = R_{p,q,\tau}(z),$$

which is (8). Since the off-diagonal entry is $\beta \neq 0$, a singular 2×2 block is nonzero and therefore has rank one. $\square$

Proposition 2.4 (Root-of-unity certificate). *Suppose $R_{p,q,\tau}$ has eight distinct roots and every root is an L th root of unity. Then, for every integer $r \geq 1$,*

$$M(P(Lr, 3)) = Z(P(Lr, 3)) = 8,$$

and the matrix $A_{Lr}(p, q, \tau)$ has nullity exactly 8.

Proof. Put $n = Lr$. Every L th root of unity is an n th root of unity, so the eight roots of $R_{p,q,\tau}$ give eight distinct singular Fourier blocks. Lemma 2.3 shows each contributes one kernel dimension, so

$$8 \leq \text{null } A_n(p, q, \tau) \leq M(P(n, 3)) \leq Z(P(n, 3)) \leq 8,$$

the last inequality being (2). Equality holds throughout. As $R_{p,q,\tau}$ has degree eight and eight distinct roots, there are no further singular blocks. $\square$

3. SELECTING SINGULAR FOURIER MODES

The next criterion converts four conjugate pairs of desired Fourier modes into parameters p, q, τ .

Theorem 3.1 (Mode-selection criterion). *Let $\theta_1, \dots, \theta_4 \in (0, \pi)$ be distinct and set $a_i = -2 \cos \theta_i$. Let e_j denote the j th elementary symmetric polynomial in $a_1, \dots, a_4$. Assume*

$$e_2 = -3 \tag{9}$$

and

$$\tau = e_1(3e_1 + e_3) - e_4 \neq 0. \tag{10}$$

Set $p = e_1$ and $q = 3e_1 + e_3$. Then

$$R_{p,q,\tau}(z) = \prod_{i=1}^4 (z^2 + a_i z + 1). \tag{11}$$

Consequently, if every $e^{i\theta_i}$ is an L th root of unity, then $M(P(Lr, 3)) = Z(P(Lr, 3)) = 8$ for every $r \geq 1$.

Proof. A direct expansion gives

$$\begin{aligned} \prod_{i=1}^4 (z^2 + a_i z + 1) &= z^8 + e_1 z^7 + (4 + e_2) z^6 + (3e_1 + e_3) z^5 + (6 + 2e_2 + e_4) z^4 \\ &\quad + (3e_1 + e_3) z^3 + (4 + e_2) z^2 + e_1 z + 1. \end{aligned}$$

Under (9) the coefficients of z^6 and z^2 equal 1 and the central coefficient becomes e_4 ; the definitions of p, q, τ give $pq - \tau = e_4$, proving (11). Since $z^2 - 2 \cos(\theta_i)z + 1 = (z - e^{i\theta_i})(z - e^{-i\theta_i})$, the roots are $e^{\pm i\theta_i}$, and the hypotheses make these eight roots distinct. Proposition 2.4 completes the proof. $\square$

Remark 3.2. The constraint $e_2 = -3$ is not an auxiliary condition introduced for convenience: within the family (6) it is exactly the coefficient constraint forced by the fixed coefficients of z^6 and z^2 in (7). The condition $\tau \neq 0$ is likewise essential, being equivalent to requiring the spoke weight $\beta = \sqrt{|\tau|}$ to be nonzero.

The following elementary symmetry explains the paired rational certificates.

Lemma 3.3 (Reflection symmetry). *For all p, q, τ ,*

$$R_{p,q,\tau}(-z) = R_{-p,-q,\tau}(z).$$

Thus negating all roots sends (p, q, τ) to $(-p, -q, \tau)$. In particular, if the roots are L th roots of unity and L is even, the reflected triple is again a certificate at modulus L ; for general L the reflected roots are $2L$ th roots of unity.

Proof. Replacing z by $-z$ in (7) negates the coefficients of z^7, z^5, z^3, z and fixes the rest; the central coefficient is unchanged because $(-p)(-q) - \tau = pq - \tau$. The statement about moduli is immediate from $(-z)^L = z^L$ when L is even. $\square$

Example 3.4 (The modulus-24 certificate). Take $\{\theta_i\} = \{\pi/4, \pi/3, 2\pi/3, 3\pi/4\}$. Then $\{a_i\} = \{-\sqrt{2}, -1, 1, \sqrt{2}\}$, so $\prod_i (t + a_i) = t^4 - 3t^2 + 2$, giving $e_1 = e_3 = 0, e_2 = -3, e_4 = 2$. Theorem 3.1 gives $(p, q, \tau) = (0, 0, -2)$ and the eight selected modes are 24th roots of unity. Equivalently $R_{0,0,-2}(z) = \Phi_3(z)\Phi_6(z)\Phi_8(z)$.

4. EXPLICIT CERTIFICATES FOR STEP SIZE THREE

Proposition 4.1. *For each row of Table 1, the polynomial $R_{p,q,\tau}$ has eight distinct roots, all of which are L th roots of unity.*

Proof. Each identity follows by multiplying the displayed cyclotomic polynomials. The factors in each row are pairwise distinct and every factor divides $z^L - 1$, so the product has eight distinct roots, all L th roots of unity. $\square$

TABLE 1. Rational root-of-unity certificates for $P(n, 3)$. The polynomial in each row is $R_{p,q,\tau}$ from (7). By Theorem 5.1 this list, together with the reflections supplied by Lemma 3.3, is complete.

L	(p, q, τ)	Cyclotomic factorization of $R_{p,q,\tau}(z)$
10	$(0, 0, -1)$	$\Phi_5(z)\Phi_{10}(z)$
24	$(0, 0, -2)$	$\Phi_3(z)\Phi_6(z)\Phi_8(z)$
40	$(1, 1, -1)$	$\Phi_5(z)\Phi_8(z)$
42	$(0, 1, -1)$	$\Phi_6(z)\Phi_7(z)$

The remaining coverage modulus uses an algebraic certificate; note that its parameters are irrational, so it lies outside the scope of Theorem 5.1.

Proposition 4.2 (The modulus-28 certificate). *Let $\alpha = 2 \cos(\pi/7)$, $p = 1 - \alpha^2$, $q = \alpha$, $\tau = -1$. Then $R_{p,q,\tau}$ has the eight distinct roots*

$$e^{\pm i\pi/14}, \quad e^{\pm i\pi/7}, \quad e^{\pm 3i\pi/7}, \quad e^{\pm 13i\pi/14},$$

all of which are 28th roots of unity.

Proof. The number $\alpha = 2 \cos(\pi/7)$ satisfies

$$\alpha^3 - \alpha^2 - 2\alpha + 1 = 0. \tag{12}$$

Let $\gamma = 2 \cos(3\pi/7)$. Since $\gamma = \alpha^3 - 3\alpha$, relation (12) gives

$$\alpha + \gamma = \alpha^2 - 1, \quad \alpha\gamma + 2 = \alpha + 1. \tag{13}$$

A direct expansion, reduced using (12), yields

$$R_{1-\alpha^2, \alpha, -1}(z) = (z^4 - \alpha z^2 + 1)(z^4 + (1 - \alpha^2)z^3 + (\alpha + 1)z^2 + (1 - \alpha^2)z + 1).$$

Let $c = 2 \cos(\pi/14)$; since $c^2 = 2 + \alpha$, $(z^2 - cz + 1)(z^2 + cz + 1) = z^4 - \alpha z^2 + 1$. Using (13),

$$(z^2 - \alpha z + 1)(z^2 - \gamma z + 1) = z^4 + (1 - \alpha^2)z^3 + (\alpha + 1)z^2 + (1 - \alpha^2)z + 1.$$

The four quadratic factors have roots $e^{\pm i\pi/14}, e^{\pm 13i\pi/14}, e^{\pm i\pi/7}, e^{\pm 3i\pi/7}$ respectively; these are eight distinct 28th roots of unity. $\square$

Theorem 4.3. *If n is a positive integer divisible by at least one of 10, 24, 28, 42, then*

$$M(P(n, 3)) = Z(P(n, 3)) = 8.$$

Proof. If $10 \mid n$, $24 \mid n$ or $42 \mid n$, apply Proposition 2.4 with the corresponding row of Table 1. If $28 \mid n$, apply Proposition 2.4 and Proposition 4.2. Every listed modulus exceeds 6, so $P(n, 3)$ is admissible. $\square$

Corollary 4.4. *The set of positive integers divisible by at least one of 10, 24, 28, 42 has natural density $1/6$.*

Proof. Let A , B , C be the sets of positive integers divisible by 10, by 24, and by 7 together with at least one of 4 or 6, respectively; then $A \cup B \cup C$ is the set in question. Their densities are $d(A) = \frac{1}{10}$, $d(B) = \frac{1}{24}$ and $d(C) = \frac{1}{7}(\frac{1}{4} + \frac{1}{6} - \frac{1}{12}) = \frac{1}{21}$. Furthermore $d(A \cap B) = \frac{1}{120}$, $d(A \cap C) = \frac{1}{105}$, $d(B \cap C) = \frac{1}{168}$ and $d(A \cap B \cap C) = \frac{1}{840}$. Inclusion–exclusion gives

$$\frac{1}{10} + \frac{1}{24} + \frac{1}{21} - \frac{1}{120} - \frac{1}{105} - \frac{1}{168} + \frac{1}{840} = \frac{140}{840} = \frac{1}{6}. \quad \square$$

Remark 4.5. The modulus 40 of Table 1 is a genuine and distinct weighting but contributes no new indices, since $40 \mid n$ implies $10 \mid n$. We retain it because Theorem 5.1 shows it is one of only four rational moduli, and because it illustrates that different weighted certificates coexist on the same arithmetic family.

5. CLASSIFICATION OF ALL RATIONAL CERTIFICATES

We now show that Table 1 is complete over $\mathbb{Q}$. This converts the table from a list of examples into a classification, and explains why no further rational certificate was found.

Theorem 5.1. *Let $p, q, \tau \in \mathbb{Q}$ with $\tau \neq 0$, and suppose $R_{p,q,\tau}$ has eight distinct roots, all of which are roots of unity. Then $R_{p,q,\tau}$ is one of the six polynomials of Table 2, and the modulus L is 10, 24, 40 or 42.*

TABLE 2. All rational certificates. The six rows form three reflection-conjugate pairs under Lemma 3.3.

Factorization	p	q	τ	L
$\Phi_5 \Phi_{10}$	0	0	−1	10
$\Phi_3 \Phi_6 \Phi_8$	0	0	−2	24
$\Phi_5 \Phi_8$	1	1	−1	40
$\Phi_8 \Phi_{10}$	−1	−1	−1	40
$\Phi_6 \Phi_7$	0	1	−1	42
$\Phi_3 \Phi_{14}$	0	−1	−1	42

Proof. Every root of $R := R_{p,q,\tau}$ is a root of unity, hence an algebraic integer; as R is monic with rational coefficients, $R \in \mathbb{Z}[z]$. Each monic irreducible factor of R over $\mathbb{Q}$ has a root of unity as a root and is therefore a cyclotomic polynomial. Since the eight roots are distinct, these factors are pairwise distinct and each occurs to the first power, so

$$R = \prod_{m \in S} \Phi_m, \quad \sum_{m \in S} \varphi(m) = 8$$

for a set S of distinct positive integers.

We first exclude $m \in \{1, 2\}$. As $\Phi_1(0) = -1$ and $\Phi_m(0) = 1$ for $m \geq 2$, while $R(0) = 1$, we get $1 \notin S$. If $2 \in S$ then $\sum_{m \in S, m \geq 3} \varphi(m) = 8 - \varphi(2) = 7$, which is impossible because $\varphi(m)$ is even for every $m \geq 3$. Hence every $m \in S$ satisfies $m \geq 3$.

Consequently S is a set of distinct integers $m \geq 3$ with $\varphi(m) \leq 8$; the possibilities are $\varphi(m) = 2$ for $m \in \{3, 4, 6\}$, $\varphi(m) = 4$ for $m \in \{5, 8, 10, 12\}$, $\varphi(m) = 6$ for $m \in \{7, 9, 14, 18\}$ and $\varphi(m) = 8$ for $m \in \{15, 16, 20, 24, 30\}$. Since $\varphi(m) \geq 2$ on this list and only three values of m give $\varphi(m) = 2$, the partitions of 8 that can occur are 8, $6 + 2$, $4 + 4$ and $4 + 2 + 2$.

Because every Φ_m with $m \geq 2$ is palindromic, so is R ; the shape of (7) therefore imposes the single further condition

$$[z^6]R = 1,$$

the equality $[z^2]R = 1$ being automatic. Checking this condition across the finitely many admissible sets S leaves exactly fourteen candidates. Of these, eight have $\tau = pq - [z^4]R = 0$ and are excluded by hypothesis, namely

$$\Phi_3\Phi_9, \Phi_6\Phi_9, \Phi_4\Phi_9, \Phi_3\Phi_{18}, \Phi_6\Phi_{18}, \Phi_4\Phi_{18}, \Phi_3\Phi_4\Phi_{12}, \Phi_4\Phi_6\Phi_{12},$$

and the remaining six are those of Table 2. Their moduli are the least common multiples of the indices appearing, namely 10, 24, 40, 40, 42, 42. $\square$

Corollary 5.2. *The set of indices covered by rational certificates is exactly $\{n : 10 \mid n \text{ or } 24 \mid n \text{ or } 42 \mid n\}$, whose natural density is $31/210$. The modulus 28 of Theorem 4.3 is therefore genuinely algebraic within the present constant-weight block-circulant class: it cannot be realized by rational parameters in (6), and it is precisely what raises the coverage density from $31/210$ to $1/6$.*

Proof. By Theorem 5.1 the rational moduli are 10, 24, 40, 42, and $40 \mid n$ implies $10 \mid n$, so the covered set is as stated. Inclusion–exclusion with $\text{lcm}(10, 24) = 120$, $\text{lcm}(10, 42) = 210$, $\text{lcm}(24, 42) = 168$ and $\text{lcm}(10, 24, 42) = 840$ gives

$$\frac{1}{10} + \frac{1}{24} + \frac{1}{42} - \frac{1}{120} - \frac{1}{210} - \frac{1}{168} + \frac{1}{840} = \frac{84+35+20-7-4-5+1}{840} = \frac{124}{840} = \frac{31}{210}.$$

Comparison with Corollary 4.4 gives $\frac{1}{6} - \frac{31}{210} = \frac{140-124}{840} = \frac{2}{105}$ for the gain contributed by the modulus 28. $\square$

6. ARITHMETIC OBSTRUCTIONS AND COMPUTATIONAL SEARCH

Theorem 5.1 settles the rational case. We now bound the algebraic case, where the parameters may lie in a number field, as they do in Proposition 4.2. Throughout this section $\zeta = \zeta_n = e^{2\pi i/n}$ and, for $k \in \mathbb{Z}$,

$$\eta_k = \zeta^k + \zeta^{-k} = 2\cos(2\pi k/n).$$

By Theorem 3.1 and Remark 3.2, a certificate at level n is exactly a choice of four distinct modes $m_1, \dots, m_4$ with $0 < m_i < n/2$ such that, writing $c_i = \eta_{m_i}$,

$$\sum_{i < l} c_i c_l = -3 \quad \text{and} \quad \tau \neq 0. \quad (14)$$

(The modes $z = \pm 1$ cannot occur: R is palindromic, so $R(z) = z^4 \tilde{R}(z + z^{-1})$ with $\tilde{R}$ of degree four, and $z = \pm 1$ corresponds to $z + z^{-1} = \pm 2$, a double root of R , contradicting distinctness.)

6.1. Degenerate configurations. We first identify a structured source of excluded solutions of $e_2 = -3$.

Proposition 6.1. *If $\tau = 0$, then*

$$R_{p,q,0}(z) = (z^2 + pz + 1)(z^6 + qz^3 + 1).$$

The six roots of the second factor are closed under multiplication by a primitive cube root of unity. Consequently, if all roots of $R_{p,q,0}$ are n th roots of unity, then $3 \mid n$.

Proof. The factorization follows by direct expansion of (7) with $\tau = 0$. The roots of $z^6 + qz^3 + 1$ are the cube roots of the two roots w_1, w_2 of $w^2 + qw + 1$, so that set is closed under multiplication by ζ_3 . If all these roots are n th roots of unity, then ζ_3 is a quotient of two n th roots of unity and hence $3 \mid n$. $\square$

6.2. A semigroup obstruction. The constraint (14) is a vanishing sum of roots of unity, and the Lam–Leung theorem [9] therefore applies. Recall its statement: if $\zeta_1 + \dots + \zeta_w = 0$ with each ζ_i an N th root of unity, then w lies in the numerical semigroup $\langle p_1, \dots, p_r \rangle$ generated by the distinct primes dividing N .

Theorem 6.2. *Suppose $A_n(p, q, \tau)$ has nullity 8 for some real p, q and some $\tau \neq 0$. Then*

$$27 \in \langle p : p \text{ prime}, p \mid n \rangle.$$

In particular:

- (1) n is not a power of 2;
- (2) if $n = p^s$ is a prime power, then $p = 3$;
- (3) n has an odd prime divisor $p \leq 23$.

Proof. Expanding $c_i c_l = (\zeta^{m_i} + \zeta^{-m_i})(\zeta^{m_l} + \zeta^{-m_l})$ into four terms, the first equation of (14) becomes

$$\sum_{i < l} \left(\zeta^{m_i + m_l} + \zeta^{-m_i - m_l} + \zeta^{m_i - m_l} + \zeta^{m_l - m_i} \right) + 3 \cdot \zeta^0 = 0,$$

a vanishing sum of $6 \cdot 4 + 3 = 27$ n th roots of unity, each with coefficient 1. By [9], 27 lies in the numerical semigroup generated by the primes dividing n .

For (i), 27 is odd and $\langle 2 \rangle$ contains only even numbers. For (ii), $27 \in \langle p \rangle$ forces $p \mid 27$, so $p = 3$. For (iii), write $27 = \sum_j a_j p_j$ with $a_j \geq 0$; since 27 is odd some odd prime p_j occurs with $a_j \geq 1$, and then $p_j \leq 27$, so $p_j \leq 23$. $\square$

6.3. Prime and semiprime levels. The prime case is already excluded by Theorem 6.2. For the more delicate case $n = 2p$ we use the following standard fact: for an odd prime p the numbers $\eta_1, \dots, \eta_{(p-1)/2}$ (formed with ζ_p) constitute a $\mathbb{Q}$ -basis of the maximal real subfield $\mathbb{Q}(\zeta_p)^+$, and

$$\sum_{k=1}^{(p-1)/2} \eta_k = \sum_{k=1}^{p-1} \zeta_p^k = -1. \quad (15)$$

See [12, Ch. 2]. We also use repeatedly the product rule $\eta_a \eta_b = \eta_{a+b} + \eta_{a-b}$, together with $\eta_{-k} = \eta_k$ and $\eta_0 = 2$.

Corollary 6.3. *If n is prime, then no constant-weight block-circulant certificate exists; that is, there are no real p, q and $\tau \neq 0$ for which $A_n(p, q, \tau)$ has nullity 8.*

Proof. If $n = p$ is prime, Theorem 6.2 gives $27 \in \langle p \rangle$, hence $p \mid 27$ and therefore $p = 3$. But $P(3, 3)$ is not an admissible generalized Petersen graph. $\square$

Theorem 6.4. *Let $n = 2p$ with p an odd prime. If a constant-weight block-circulant certificate exists at level n , then $p = 5$ and $n = 10$.*

Proof. The admissible modes are distinct elements of $\{1, \dots, p-1\}$ and $c_m = 2\cos(\pi m/p)$. Since $\zeta_{2p} = -\zeta_p^{(p+1)/2}$, we have $\zeta_{2p}^m = (-1)^m \zeta_p^{m(p+1)/2}$, so

$$c_m = \delta_m \eta_{j(m)}, \quad \delta_m = (-1)^m, \quad j(m) \equiv \frac{p+1}{2}m \pmod{p},$$

with $j(m)$ folded into $\{1, \dots, (p-1)/2\}$ and η formed with ζ_p . The map $m \mapsto (j(m), \delta_m)$ is a bijection from $\{1, \dots, p-1\}$ onto $\{1, \dots, \frac{p-1}{2}\} \times \{\pm 1\}$: indeed $j(p-m) = -j(m)$ folds to $j(m)$, while m and $p-m$ have opposite parity.

Write $c_i = \delta_i \eta_{j_i}$ for the four chosen modes. Then $c_i c_l = \delta_i \delta_l (\eta_{j_i+j_l} + \eta_{j_i-j_l})$, where $j_i + j_l \in \{2, \dots, p-1\}$ is never $\equiv 0$, and $j_i - j_l \equiv 0$ exactly when $j_i = j_l$. Using $\eta_0 = 2 = -2 \sum_k \eta_k$ from (15) and expanding in the basis $\{\eta_k\}$, write $\sum_{i < l} c_i c_l = \sum_k A_k \eta_k$. By (15) and the $\mathbb{Q}$ -linear independence of the η_k , the constraint forces $A_k = 3$ for all k , hence

$$\sum_k A_k = \frac{3(p-1)}{2}. \quad (16)$$

A pair with $j_i \neq j_l$ contributes $2\delta_i \delta_l$ to $\sum_k A_k$; a pair with $j_i = j_l$ has $\delta_i = -\delta_l$ and contributes $(-1)(1 - 2 \cdot \frac{p-1}{2}) = p-2$. Let Z be the number of pairs with $j_i = j_l$ and $D = \sum \delta_i \delta_l$ over the remaining pairs; then (16) becomes

$$Z(p-2) + 2D = \frac{3(p-1)}{2}.$$

Since each value of j is taken by at most two modes, $Z \in \{0, 1, 2\}$.

If $Z = 2$ the modes are $(j, +), (j, -), (j', +), (j', -)$ and the four cross pairs carry signs $+, -, -, +$, so $D = 0$ and $2(p-2) = \frac{3(p-1)}{2}$, giving $p = 5$.

If $Z = 1$ the modes are $(j, +), (j, -), (j_2, \delta_2), (j_3, \delta_3)$, whose five cross pairs carry signs $\delta_2, \delta_3, -\delta_2, -\delta_3, \delta_2 \delta_3$, so $D = \delta_2 \delta_3 = \pm 1$ and $(p-2) \pm 2 = \frac{3(p-1)}{2}$, giving $p = 3$ or $p = -5$; neither yields an admissible $n \geq 7$.

If $Z = 0$ then $D = \sum_{i < l} \delta_i \delta_l = \frac{(\sum_i \delta_i)^2 - 4}{2} \in \{-2, 0, 6\}$, and $2D = \frac{3(p-1)}{2}$ gives $p = 9$, $p = 1$ or a negative value; none is an odd prime.

Hence $p = 5$, and the modulus 10 of Table 1 is realized. $\square$

6.4. All prime powers are excluded over the reals.

Theorem 6.5. *No admissible prime-power level n supports a constant-weight block-circulant nullity-eight certificate of this class.*

Proof. Nullity eight requires eight distinct n th roots of R , none equal to ± 1 : reciprocal polynomials of even degree have even root multiplicity at either endpoint. Write these roots as $\zeta^{\pm m_i}$ for four distinct modes $0 < m_i < n/2$. Their traces $c_i = \zeta^{m_i} + \zeta^{-m_i}$ satisfy $\sum_{i < j} c_i c_j = -3$.

Reduce all exponents modulo n and form

$$F(X) = 3 + \sum_{i < j} (X^{m_i+m_j} + X^{m_i-m_j} + X^{-m_i+m_j} + X^{-m_i-m_j}).$$

Then $F(\zeta) = 0$, $\deg F < n$, $F(1) = 27$, and $[X^0]F = 3$, since no pair of distinct modes agrees up to sign. For $n = \ell^s$ with ℓ prime, write $h = \ell^{s-1}$. Divisibility by $\Phi_n = 1 + X^h + \dots + X^{(\ell-1)h}$ shows that coefficients in each class $r, r+h, \dots, r+(\ell-1)h$ are equal: the quotient has degree less than h . Therefore $\ell \mid F(1) = 27$ and $\ell = 3$.

It remains to treat $n = 3^s$ and $h = n/3$. Let $A = \{\pm m_i : 1 \leq i \leq 4\} \subset \mathbb{Z}/n\mathbb{Z}$. It is an eight-element symmetric set excluding zero. Coefficient equality gives $[X^h]F = 3$. Let $D(h)$

count ordered pairs $(a, b) \in A^2$ with $a - b = h$. Negating b instead counts sums. Distinct-element unordered pairs summing to h are counted by $[X^h]F$, contributing 6 ordered pairs. The only possible diagonal contribution solves $2a = h$, namely $a = -h$. Thus $D(h) = 6 + \delta$, where $\delta = \mathbf{1}_{h \in A}$.

Partition the group into cosets of $H = \{0, h, -h\}$. The only self-inverse coset is H because the quotient has odd order; it has 2δ elements of A . Write $r_j \in \{0, 1, 2, 3\}$ for the common occupancy of a pair of other inverse cosets. Then $\delta + \sum_j r_j = 4$. A subset of size r in a three-element coset contributes $\binom{r}{2}$ ordered differences equal to h (check $r = 0, 1, 2, 3$). Consequently

$$D(h) = \delta + 2 \sum_j \binom{r_j}{2}, \quad \sum_j \binom{r_j}{2} = 3.$$

If all $r_j \leq 2$, three occupancies of size two would be needed, violating $\sum r_j \leq 4$. There is therefore exactly one occupancy three, and all others are at most one. Hence A consists of two full inverse H -cosets and a remaining inverse pair.

The polynomial of the corresponding roots is

$$R(z) = (z^6 - tz^3 + 1)(z^2 - cz + 1).$$

Comparison gives $p = -c$, $q = -t$, and $pq - \tau = ct$, hence $\tau = 0$, contradicting admissibility. $\square$

This obstruction applies to the constant-weight class only. It imposes no such conclusion on arbitrary vertex-dependent weightings or on $M(P(n, 3))$.

6.5. Complete exact search. For each n , choose a prime $\ell \equiv 1 \pmod{n}$ and an element w of exact order n in $\mathbb{F}_\ell$. The values $d_m = w^m + w^{-m}$ are distinct for $0 < m < n/2$: equality forces equality of the unordered roots of $t^2 - d_m t + 1$. For each triple $i < j < k$, a fourth mode satisfies the necessary equation

$$(d_i + d_j + d_k)d_l = -3 - d_i d_j - d_i d_k - d_j d_k.$$

If the coefficient on the left is nonzero, solve and look up $l > k$. If it is zero, discard only when the right side is nonzero; otherwise enumerate all $l > k$. The homomorphism $\mathbb{Z}[\zeta_n] \rightarrow \mathbb{F}_\ell$ sending ζ_n to w ensures that every exact solution survives this step.

Configurations containing a three-mode subset whose six roots form two complete cube-root orbits are discarded exactly: their three traces have sum zero and pair sum -3 , and adjoining any fourth trace forces $\tau = 0$. All other retained candidates are checked for $e_2 = -3$ and $\tau \neq 0$ in $\mathbb{Z}[z]/(\Phi_n)$.

Theorem 6.6 (Computer-assisted finite classification). *For $7 \leq n \leq 420$, the primitive levels of admissible real constant-weight certificates are exactly*

$$10, 24, 28, 40, 42, 60, 84.$$

In particular, no additional divisibility family has base level at most 420.

Computation and reproducibility. The complete sieve above is implemented in `exact_candidates.c`; exact cyclotomic verification is implemented in `exact_search.py`. Primitive level means $\text{lcm}_i(n/\gcd(n, m_i))$. The full run retains 361 modular candidates and records the exact primitive certificates in `exact_search_all_7_420.json`. The distinct primitive levels in that file are precisely the displayed ones. Levels 40, 60, 84 add no new divisibility families.

An additional run omitting multiples of 10, 24, 28, 42 retains only six modular candidates, all rejected by exact verification. The generator was tested against exhaustive exact cyclotomic

enumeration for $9 \leq n \leq 30$. Code review checked the zero-denominator case, inversion batching, degeneracy shortcut and signed 64-bit product bounds through 420. This is a bounded computational result, not a global classification. $\square$

To reproduce with Python 3.10 or later, SymPy, and a C11 compiler:

```
gcc -O3 -std=c11 exact_candidates.c -o exact_candidates
python exact_search.py ./exact_candidates 7 420 \
  --output exact_search_all_7_420.json
python verify_signed_classification.py
python test_exact_search.py ./exact_candidates
```

On Windows, use the appropriate executable suffix. JSON parameter coefficients are in descending powers of z , evaluated at ζ_n after reduction modulo Φ_n .

7. A SIGNED CERTIFICATE FOR GENERAL STEP SIZE

A particularly simple specialization of Lemma 2.1 works for arbitrary k . Define

$$S_{n,k} = \begin{pmatrix} C_n + C_n^{-1} & I_n \\ I_n & -(C_n^k + C_n^{-k}) \end{pmatrix} \in \mathcal{S}(P(n, k)), \quad (17)$$

and $Q_k(z) = z^{2k+2} + z^{2k} + z^{k+1} + z^2 + 1$.

Theorem 7.1 (Signed-circulant certificate). *Suppose $Q_k(z)$ divides $z^L - 1$ in $\mathbb{Z}[z]$. Then, for every integer $r \geq 1$, $M(P(Lr, k)) = Z(P(Lr, k)) = 2k + 2$, and $S_{Lr,k}$ has nullity exactly $2k + 2$.*

Proof. Because $\deg Q_k = 2k + 2$, the divisibility assumption forces $L \geq 2k + 2$, so $P(Lr, k)$ is admissible. For z an n th root of unity, the corresponding Fourier block of (17) is

$$\begin{pmatrix} z + z^{-1} & 1 \\ 1 & -(z^k + z^{-k}) \end{pmatrix},$$

with determinant

$$-(z + z^{-1})(z^k + z^{-k}) - 1 = -z^{-(k+1)}Q_k(z). \quad (18)$$

Since $z^L - 1$ is squarefree in characteristic zero, so is every divisor of it; thus Q_k has exactly $2k + 2$ distinct roots, all L th roots of unity. If $n = Lr$ these are n th roots of unity, so (18) gives $2k + 2$ singular Fourier blocks, each nonzero because its off-diagonal entries equal 1 and hence each contributing one kernel dimension. Therefore

$$2k + 2 \leq \text{null } S_{n,k} \leq M(P(n, k)) \leq Z(P(n, k)) \leq 2k + 2,$$

the last inequality being (2). The degree of Q_k shows there are no additional singular blocks. $\square$

For $k = 3$ we get $Q_3(z) = z^8 + z^6 + z^4 + z^2 + 1 = \Phi_5(z)\Phi_{10}(z)$, recovering the modulus-10 certificate.

Corollary 7.2. *For every integer $r \geq 1$, $M(P(120r, 7)) = Z(P(120r, 7)) = 16$.*

Proof. A direct factorization gives $Q_7(z) = z^{16} + z^{14} + z^8 + z^2 + 1 = \Phi_5(z)\Phi_{10}(z)\Phi_{24}(z)$, and $\text{lcm}(5, 10, 24) = 120$. Apply Theorem 7.1. $\square$

Remark 7.3. For $k = 2$ one has $Q_2 = \Phi_5\Phi_6$, so Theorem 7.1 gives

$$M(P(30r, 2)) = Z(P(30r, 2)) = 6.$$

This is contained in the stronger family $M(P(15r, 2)) = Z(P(15r, 2)) = 6$ of [2] and therefore serves only as a consistency check on the signed construction.

Write $\chi_m(k)$ for the indicator of the residue classes in Table 3.

TABLE 3. All possible distinct cyclotomic factors of Q_k .

m	$k \pmod{m}$ with $\chi_m(k) = 1$	$\varphi(m)$
5	2,3	4
6	2,4	2
10	3,7	4
24	7,17	8

7.1. The signed classification and exact nullity.

Theorem 7.4. *For every admissible n, k ,*

$$\text{null } S_{n,k} = 4\mathbf{1}_{5|n}\chi_5(k) + 2\mathbf{1}_{6|n}\chi_6(k) \\ + 4\mathbf{1}_{10|n}\chi_{10}(k) + 8\mathbf{1}_{24|n}\chi_{24}(k).$$

In particular, this nullity is at most 6 for even k , and at most 16 for odd k .

Theorem 7.5. *For $k \geq 1$, the polynomial Q_k divides $z^L - 1$ for some positive integer L if and only if $k \in \{2, 3, 7\}$. The minimal L is respectively 30, 10, 120.*

Lemma 7.6. *Every root-of-unity zero of Q_k has order dividing 10 or 24.*

Proof. The classical classification of five-term vanishing sums of roots of unity, counting multiplicities, says that the terms form a rotated regular pentagon or a rotated regular triangle together with an antipodal pair. This follows from the minimal relations of weights at most five in [10, Theorem 3 and Table 1]: only weights 2, 3, 5 occur, as regular polygons, and a nonminimal weight-five relation must split as $2 + 3$.

Let ζ have order N and satisfy $Q_k(\zeta) = 0$. Label the exponent occurrences $A = 0$, $B = 2$, $C = k + 1$, $D = 2k$, $E = 2k + 2$; these are labels even when exponents coincide for $k = 1$. In the pentagon case the ratio of terms B and A is a fifth root of unity, giving $N \mid 10$.

In the triangle case, the cube of the ratio of any two triangle terms is one. The six triples $ABC, ABD, ABE, ADE, BDE, CDE$ contain an exponent difference of 2, giving $N \mid 6$. The triple ACD gives $N \mid 3(k + 1)$ and $N \mid 6k$, whose integer linear combinations give $N \mid 6$. The same argument for BCE uses $3(k - 1)$ and $6k$.

For ACE the triangle gives $N \mid 3(k + 1)$, and its complementary antipodal pair BD gives $N \mid 4(k - 1)$. Taking their integer linear combination gives $N \mid 24$. For BCD the two divisibilities are $N \mid 3(k - 1)$ and $N \mid 4(k + 1)$, with the same conclusion. $\square$

Proof of the two signed-classification theorems. By Lemma 7.6, the only cyclotomic indices to consider are

$$1, 2, 3, 4, 5, 6, 8, 10, 12, 24.$$

For each fixed m , the remainder of Q_k modulo Φ_m depends only on $k \pmod{m}$. Reducing the five monomials for these finitely many residues gives Table 3; indices 1, 2, 3, 4, 8, 12 never occur. The accompanying exact verification program `verify_signed_classification.py` reproduces every remainder in exact integer polynomial arithmetic. The calculation can also be checked from

$$Q_k(z) = z^{k+1}((z + z^{-1})(z^k + z^{-k}) + 1).$$

The Fourier blocks of $S_{n,k}$, for $z^n = 1$, have determinant $-z^{-(k+1)}Q_k(z)$. Their off-diagonal entries are 1, so every singular block contributes exactly one to nullity. A factor Φ_m contributes $\varphi(m)$ modes exactly when $m \mid n$. The table now proves Theorem 7.4; possible multiplicities of polynomial factors are irrelevant to this count.

For Theorem 7.5, a divisor of $z^L - 1$ is squarefree. When k is even the only possible factors are Φ_5, Φ_6 , of total degree 6, so $2k + 2 \leq 6$ forces $k = 2$. When k is odd the possible factors have total degree $4 + 4 + 8 = 16$, forcing $k \leq 7$. Among 1, 3, 5, 7 the table excludes 1, 5. Finally direct multiplication gives

$$Q_2 = \Phi_5 \Phi_6, \quad Q_3 = \Phi_5 \Phi_{10}, \quad Q_7 = \Phi_5 \Phi_{10} \Phi_{24}.$$

These are products of distinct factors. Their index least common multiples are 30, 10, 120, proving sufficiency and minimality. $\square$

8. CERTIFICATES OVER FINITE FIELDS

We now change the coefficient field. The elementary zero-forcing kernel argument needs neither an ordering of the field nor symmetry of the matrix. For clarity, $M(G)$ continues to denote maximum nullity over real symmetric matrices only.

Lemma 8.1 (Kernel restriction over any field). *Let K be any field and let $T = (t_{ij})$ be a matrix indexed by $V(G)$ such that, for distinct i, j , $t_{ij} \neq 0$ if and only if $ij \in E(G)$. Then $\text{null}_K T \leq Z(G)$, even if T is not symmetric.*

Proof. Let B be a zero forcing set and let $x \in \ker T$ vanish on B . Whenever a blue vertex i forces its unique white neighbor j , all coordinates at i and its other neighbors are already zero. Row i of $Tx = 0$ reduces to $t_{ij}x_j = 0$. The coefficient is nonzero, hence $x_j = 0$. Induction through a forcing sequence shows $x = 0$. Thus restriction $\ker T \rightarrow K^B$ is injective, and $\dim_K \ker T \leq |B|$. Minimize over B . $\square$

Theorem 8.2 (Polynomial gcd certificate). *Let $n \geq 7$, let K be a field, and let $p, b, q, d, \tau \in K$ with $b\delta\tau \neq 0$. For the cyclic shift C_n , set*

$$U = C_n + pI + bC_n^{-1}, \quad V = C_n^3 + qI + dC_n^{-3}, \quad T_n = \begin{pmatrix} U & I \\ \tau I & V \end{pmatrix}.$$

Define the monic degree-eight polynomial

$$\begin{aligned} R_{p,b,q,d,\tau}(z) &= (z^2 + pz + b)(z^6 + qz^3 + d) - \tau z^4 \\ &= z^8 + pz^7 + bz^6 + qz^5 + (pq - \tau)z^4 + bqz^3 + dz^2 + pdz + bd. \end{aligned}$$

Then T_n has exactly the off-diagonal graph pattern $P(n, 3)$, and

$$\text{null}_K T_n = \deg \gcd(R_{p,b,q,d,\tau}(z), z^n - 1).$$

In particular, if $R_{p,b,q,d,\tau} \mid z^L - 1$, then $Z(P(Lr, 3)) = 8$ for every positive integer r .

Proof. The shifts C_n, C_n^{-1} and C_n^3, C_n^{-3} occupy the two distinct outer and inner neighbor positions since $n \geq 7$; their coefficients and both spoke coefficients are nonzero. Thus the graph pattern is exact. Writing a kernel vector as (x, y) , the first block equation gives $y = -Ux$; the second gives $(VU - \tau I)x = 0$. Multiplication by the invertible matrix C_n^4 turns this operator into $R_{p,b,q,d,\tau}(C_n)$.

Identify the cyclic vector space with $K[z]/(f)$, where $f = z^n - 1$, so that C_n acts as multiplication by z . Write $g = \gcd(R, f)$, $f = gf_1$, and $R = gR_1$. A residue class h is killed by R precisely when $f \mid Rh$, equivalently $f_1 \mid R_1h$. Since $\gcd(f_1, R_1) = 1$, this is equivalent to $f_1 \mid h$. The killed classes therefore form a space of dimension $\deg g$. This proves the formula without assuming that f is squarefree, so it also covers characteristics dividing n .

If $R \mid z^L - 1$, then $L \geq 8$ by degrees and $R \mid z^{Lr} - 1$. The gcd formula gives nullity eight for T_{Lr} . Apply Lemma 8.1 and the upper bound (2). $\square$

When $b = d = 1$, multiplication of the bottom block row by τ^{-1} yields the symmetric finite-field matrix $\begin{pmatrix} U & I \\ I & \tau^{-1}V \end{pmatrix}$. Its kernel is unchanged. Symmetry is thus available in every row below except $L = 16$, but Lemma 8.1 also validates that row.

Corollary 8.3 (A binary certificate). *For every positive integer r , $Z(P(17r, 3)) = 8$.*

Proof. Over $\mathbb{F}_2$ take $(p, b, q, d, \tau) = (1, 1, 0, 1, 1)$. Direct multiplication in $\mathbb{F}_2[z]$ gives

$$\begin{aligned} z^{17} - 1 &= (z^8 + z^7 + z^6 + z^4 + z^2 + z + 1) \\ &\quad \cdot (z^9 + z^8 + z^6 + z^3 + z + 1). \end{aligned}$$

The first factor is $R_{1,1,0,1,1}$, so Theorem 8.2 applies, including for even r . $\square$

TABLE 4. Exact finite-field certificates. Each row means $R_{p,b,q,d,\tau} \mid z^L - 1$ in $\mathbb{F}_\ell[z]$; parameters are represented by integers modulo the prime ℓ .

L	ℓ	p	b	q	d	τ
16	17	9	6	1	12	11
17	2	1	1	0	1	1
18	19	4	1	8	1	11
19	229	15	1	12	1	84
21	41	28	1	22	1	33
22	23	8	1	21	1	14
23	461	111	1	234	1	146
25	499	364	1	355	1	26
26	5	2	1	4	1	2
27	271	45	1	230	1	181
29	233	46	1	188	1	61
31	743	619	1	574	1	699

Proof of Theorem 1.5. For every row of Table 4, polynomial division over $\mathbb{F}_\ell$ gives zero remainder and $bd\tau \neq 0$. These finitely many exact identities can be checked directly from the seven displayed integers, or reproduced with `round2/verify_finite_fields.py`. Theorem 8.2 gives the assertion for every multiple of each base level. As a separate computational check, direct modular Gaussian elimination and graph-pattern checks give nullity eight at $n = L$ and $n = 2L$ for each row. The checks at $n = 34$ over $\mathbb{F}_2$ and $n = 130$ over $\mathbb{F}_5$ also cover inseparable cases. The matrix checks are corroboration; divisibility and the theorem prove all multiples. $\square$

Corollary 8.4 (Coverage of the combined families). *The union of the original real families and the finite-field families in Table 4 has natural density*

$$\frac{6933083}{14969435} = 0.463149277\dots$$

At every index in this union, $Z(P(n, 3)) = 8$.

Proof. Remove the original base level 42, which is a multiple of 21. The remaining bases are

$$\mathcal{L} = \{10, 16, 17, 18, 19, 21, 22, 23, 24, 25, 26, 27, 28, 29, 31\}.$$

A common multiple of a subset $J \subseteq \mathcal{L}$ has density $1/\text{lcm}(J)$. Inclusion–exclusion gives

$$\sum_{\emptyset \neq J \subseteq \mathcal{L}} \frac{(-1)^{|J|+1}}{\text{lcm}(J)} = \frac{6933083}{14969435}.$$

This is a finite rational calculation, reproduced in the supplementary coverage record. The zero-forcing conclusion follows from the two families of certificates. This density describes a proved subset of indices, not the density of all indices satisfying the conjecture. $\square$

These finite-field identities are compatible with the real prime-power obstruction. Reduction modulo a prime may lower rank, so a finite-field kernel need not lift to a real kernel. Consequently the newly covered families establish ordinary zero forcing, without extending the asserted real maximum-nullity families.

9. SMALL ZERO FORCING NUMBERS FOR GENERALIZED PETERSEN GRAPHS

Krishnan [8] computed $Z(P(n, 3))$ exhaustively for $7 \leq n \leq 20$. We independently implemented the same finite search and reproduced those values; the original exhaustive computation through $n = 27$ is retained in Table 5. The anchored extension below reaches $n = 64$. For each n , subsets are tested in increasing cardinality and the colour-change rule is iterated to closure. The results are collected in Table 5.

TABLE 5. Zero forcing numbers of $P(n, 3)$, by exhaustive search over all vertex subsets. For every n in the range $13 \leq n \leq 27$ a minimum zero forcing set is given by eight consecutive outer vertices.

n	7	8	9	10	11	12	13	14	15	16	17
$Z(P(n, 3))$	6	6	6	8	7	7	8	8	8	8	8
n	18	19	20	21	22	23	24	25	26	27	
$Z(P(n, 3))$	8	8	8	8	8	8	8	8	8	8	

The entries through $n = 20$ agree with Krishnan [8]. Only $n = 12$ contradicts the published range $n \geq 12$ of Rashidi et al.; the value $Z(P(11, 3)) = 7$ lies outside that claimed range. The original extension verified $Z(P(n, 3)) = 8$ also for $21 \leq n \leq 27$. In particular, the prime values $n = 13, 17, 19, 23$ satisfy the zero-forcing conjecture even though Corollary 6.3 shows that no constant-weight block-circulant real maximum-nullity certificate of the present type exists at those levels. The finite-field certificates above now cover all multiples of 17, 19, and 23 without contradicting that obstruction.

9.1. An exhaustive first-force reduction.

Lemma 9.1 (First-force anchoring). *For $n \geq 7$ and $3 \leq r < 2n$, $P(n, 3)$ has a zero forcing set of size at most r if and only if it has a size- r zero forcing set containing one of the following seeds (indices are modulo n):*

$$\{u_0, u_{-1}, u_1\}, \quad \{u_0, u_1, v_0\}, \quad \{v_0, v_{-3}, v_3\}, \quad \{v_0, v_3, u_0\}.$$

Proof. Enlarge a forcing set of size at most r to size r , which preserves forcing. Because $r < 2n$, a first force $x \rightarrow y$ must occur. The vertex x and its two neighbors other than y are initially blue. Rotate indices so that x has index zero. If x is outer, then y is either its spoke neighbor, giving the first seed, or an outer neighbor, giving the second seed after reflection if needed. If x is inner, the analogous alternatives give the third and fourth seeds. Rotation and simultaneous

reflection of both cycles are graph automorphisms. Conversely, every size- r set tested with any seed is an actual vertex set of the graph. $\square$

For $r = 7$ the lemma bounds the number of tested sets by $4\binom{2n-3}{4}$, instead of $\binom{2n}{7}$. Overlap among seeds does not affect exhaustiveness. A queue-based closure takes linear time on a cubic graph: enqueue each initially blue or newly blue vertex, and enqueue any blue neighbor enabled when a vertex turns blue. Each vertex's white-neighbor count decreases monotonically, so it crosses one at most once. There are therefore only $O(n)$ queue events. The resulting exhaustive algorithm uses $O(n^5)$ time and $O(n)$ working space.

Theorem 9.2 (Finite exhaustive verification). *For every integer $13 \leq n \leq 64$, $Z(P(n, 3)) = 8$.*

Computation and checks. The program `round2/direct_zf_anchor.c` exhausts the four anchored families for every n in the stated interval and finds no forcing set of size seven. Lemma 9.1 excludes all smaller sets as well. The eight consecutive outer vertices force in every tested graph, independently confirming the upper bound.

Anchored and unrestricted enumeration agree for $7 \leq n \leq 12$ and $3 \leq r \leq 7$ (thirty decisions). The queue closure agrees with an independent rescan closure for every initial subset at $n = 7, 8, 9$ (344,064 subsets). The complete run logs are `round2/direct_zf_results_13_40.txt` and `round2/direct_zf_results_41_64.txt`. The implementation uses unsigned 128-bit masks and supports $n \leq 64$; this computational statement is confined to the recorded interval. $\square$

To reproduce with a compiler supporting unsigned 128-bit integers:

```
gcc -std=c11 -O3 round2/direct_zf_anchor.c -o direct_zf_anchor
./direct_zf_anchor 7 12 selftest
./direct_zf_anchor 13 64
```

10. DISCUSSION AND OPEN PROBLEMS

The real and finite-field arguments provide different conclusions. The four original real divisibility families have $M(P(n, 3)) = Z(P(n, 3)) = 8$ and density $1/6$. The additional families in Table 4 establish $Z(P(n, 3)) = 8$ by changing characteristic. They include prime and prime-power base levels excluded from the real constant-weight symmetric construction. Thus an obstruction to one matrix class is not an obstruction to the graph's zero-forcing conjecture.

The rational classification is complete; the real algebraic classification is complete only through 420. The signed-polynomial question is resolved for every k , and the direct computation verifies $13 \leq n \leq 64$. None of these results supplies the uniform lower bound for all $n \geq 13$. The finite-field route suggests replacing a search for real Fourier modes by a search for degree-eight divisors over suitable characteristics, but existence for every graph order has not been proved.

Question 10.1. Does every $n \geq 13$ admit a field and nonzero b, d, τ for which the polynomial in Theorem 8.2 divides $z^n - 1$? More broadly, can graph-pattern matrices over varying fields prove the uniform zero-forcing conjecture?

Question 10.2. Is $M(P(n, 3)) = 8$ over the reals for every $n \geq 13$? At prime-power levels, Theorem 6.5 requires any real symmetric witnessing matrix to leave the constant-weight block-circulant class.

Question 10.3. Do real edge weightings with a nontrivial period produce additional maximum-nullity families? For period $m \mid n$, the corresponding Fourier blocks have size $2m$, so their determinants allow more parameters.

Question 10.4. Are 10, 24, 28, 42 the only base moduli producing distinct divisibility families in the real constant-weight class? The exact classification through 420 is finite evidence; the arithmetic obstructions do not supply a global bound on admissible levels.

Question 10.5. Can the four first-force anchors be combined with a uniform obstruction to forcing for every choice of four additional vertices? A proof would need to cover arbitrary cyclic gaps, beyond the finite range tested here.

REPRODUCIBILITY

The original supplementary scripts are preserved. Their eighteen symbolic checks, complete rational enumeration, and exhaustive zero-forcing results for $7 \leq n \leq 27$ have been rerun. All twenty-one supplied forcing witnesses were also checked by an independent set-based closure implementation. The extended supplement includes exact signed-polynomial residue checks, the complete real modular sieve and cyclotomic verification through 420, finite-field polynomial and matrix checks, and the anchored search through 64. The proofs specify the mathematical scope of each computation; no theorem relies on numerical optimization. In particular, finite-field certificates are exact identities, while a finite zero-forcing search gives no conclusion outside its tested range.

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REFERENCES

- [1] AIM Minimum Rank–Special Graphs Work Group, *Zero forcing sets and the minimum rank of graphs*, Linear Algebra Appl. **428** (2008), no. 7, 1628–1648, <https://doi.org/10.1016/j.laa.2007.10.009>.
- [2] J. S. Alameda, E. Curl, A. Grez, L. Hogben, O. Kingston, A. Schulte, D. Young, and M. Young, *Families of graphs with maximum nullity equal to zero forcing number*, Spec. Matrices **6** (2018), no. 1, 56–67, <https://doi.org/10.1515/spma-2018-0006>.
- [3] F. Barioli, W. Barrett, S. M. Fallat, H. T. Hall, L. Hogben, B. L. Shader, P. van den Driessche, and H. van der Holst, *Zero forcing parameters and minimum rank problems*, Linear Algebra Appl. **433** (2010), no. 2, 401–411, <https://doi.org/10.1016/j.laa.2010.03.008>.
- [4] P. J. Davis, *Circulant Matrices*, 2nd ed., AMS Chelsea Publishing, Providence, RI, 1994.
- [5] L. Duong, B. K. Kroschel, M. Riddell, K. N. Vander Meulen, and A. Van Tuyl, *Maximum nullity and zero forcing of circulant graphs*, Spec. Matrices **8** (2020), no. 1, 221–234, <https://doi.org/10.1515/spma-2020-0106>.
- [6] S. M. Fallat and L. Hogben, *The minimum rank of symmetric matrices described by a graph: A survey*, Linear Algebra Appl. **426** (2007), no. 2–3, 558–582, <https://doi.org/10.1016/j.laa.2007.05.036>.
- [7] R. Gera and P. Stănică, *The spectrum of generalized Petersen graphs*, Australas. J. Combin. **49** (2011), 39–45.
- [8] A. Krishnan, *A correction to the zero forcing number of the generalized Petersen graphs $P(n, 3)$* , arXiv:2607.19412 (2026), <https://doi.org/10.48550/arXiv.2607.19412>.
- [9] T. Y. Lam and K. H. Leung, *On vanishing sums of roots of unity*, J. Algebra **224** (2000), no. 1, 91–109, <https://doi.org/10.1006/jabr.1999.8089>.
- [10] B. Poonen and M. Rubinstein, *The number of intersection points made by the diagonals of a regular polygon*, SIAM J. Discrete Math. **11** (1998), no. 1, <https://doi.org/10.1137/S0895480195281246>. Author version: <https://math.mit.edu/~poonen/papers/ngon.pdf>.

- [11] S. Rashidi, N. Shajareh Poursalavati, and M. Tavakkoli, *Computing the zero forcing number for generalized Petersen graphs*, J. Algebra Comb. Discrete Struct. Appl. **7** (2020), no. 2, 183–193,
<https://doi.org/10.13069/jacodesmath.729465>.
- [12] L. C. Washington, *Introduction to Cyclotomic Fields*, 2nd ed., Graduate Texts in Mathematics **83**, Springer, New York, 1997.
- [13] M. E. Watkins, *A theorem on Tait colorings with an application to the generalized Petersen graphs*, J. Combin. Theory **6** (1969), no. 2, 152–164,
[https://doi.org/10.1016/S0021-9800\(69\)80116-X](https://doi.org/10.1016/S0021-9800(69)80116-X).
- [14] D. Young, *Techniques for determining equality of the maximum nullity and the zero forcing number of a graph*, Electron. J. Linear Algebra **37** (2021), 295–315,
<https://doi.org/10.13001/ela.2021.4967>.

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